TIME SERIES ANALYSIS PROJECT REPORT

1. Introduction

Time Series Analysis is a statistical technique used to study data collected over a period of time.

It helps us understand past patterns, identify trends and seasonal changes, and predict future values.

In this project, we use Python to analyse monthly airline passenger data from 1949 to 1960.

We apply different time series techniques, including trend analysis, seasonality analysis, stationarity testing, differencing, decomposition, ARIMA forecasting, and exponential smoothing.

2. Objectives

The main objectives of this project are:

1. To understand the basics of time series analysis.
2. To convert date columns into datetime format.
3. To visualize time series data using graphs.
4. To identify trends and seasonal patterns.
5. To check whether data is stationary or non-stationary.
6. To perform differencing and moving average calculations.
7. To study ACF and PACF plots.
8. To apply ARIMA forecasting.
9. To perform Simple Exponential Smoothing.
10. To apply the Holt-Winters forecasting method.
11. To predict future airline passenger numbers.

12. Software and Tools Used

- Programming Language: Python
- Code Editor: Visual Studio Code
- Libraries: Pandas, NumPy, Matplotlib, Seaborn, Statsmodels and Scikit-learn
- Dataset: Airline Passengers CSV file
- Output: Graphs, CSV files and forecasting results

4. Dataset Description

The dataset used in this project is Airline Passengers.

- Dataset name: airline_passengers.csv
- Time period: January 1949 to December 1960
- Total records: 144 monthly observations

- Date column: Month
- Data column: Passengers

The dataset contains monthly information about the number of airline passengers.

5. Methodology

The following steps were performed in this project:

1. Load the CSV dataset using Pandas.
2. Convert the date column into datetime format.
3. Set and analyse the time series data.
4. Visualize monthly passenger numbers.
5. Identify trend and seasonal patterns.
6. Test stationarity using the Augmented Dickey-Fuller (ADF) test.
7. Apply differencing to make the data stationary.
8. Calculate moving averages and EWMA.
9. Perform time series decomposition.
10. Analyse ACF and PACF plots.
11. Apply forecasting models.
12. Save graphs and forecast results.
13. Practical Programs

The project contains 13 practical programs:

1. Datetime Conversion
2. Time Series Visualization
3. Trend Analysis
4. Seasonality Analysis
5. Stationarity Analysis
6. Augmented Dickey-Fuller Test
7. Differencing
8. Moving Average and EWMA
9. Time Series Decomposition
10. ACF and PACF
11. ARIMA Forecasting
12. Simple Exponential Smoothing
13. Holt-Winters Forecasting
14. Forecasting Models

7.1 ARIMA Model

ARIMA stands for AutoRegressive Integrated Moving Average.

It is a forecasting model used for time series data. It uses past observations and errors to predict future values.

In this project, ARIMA is used to forecast airline passenger numbers for the next 12 months.

Output: arima_forecast_results.csv

7.2 Simple Exponential Smoothing

Simple Exponential Smoothing is a forecasting method that gives more importance to recent observations.

It is useful for data without a clear trend or seasonal pattern.

In this project, it is used to forecast the next 12 months of airline passengers.

Output: simple_exponential_smoothing_results.csv

7.3 Holt-Winters Method

Holt-Winters is a forecasting method that considers level, trend and seasonality.

It is useful for time series data containing seasonal patterns.

In this project, it is applied to forecast future airline passenger numbers.

Output: holt_winters_results.csv

8. Output Files

The project generates the following outputs:

- Graph images saved in outputs/plots/.
- ARIMA forecast results saved in CSV format.
- Simple Exponential Smoothing results saved in CSV format.
- Holt-Winters results saved in CSV format.

9. Result Analysis

The airline passenger dataset shows changes in passenger numbers over time.

The analysis helps identify:

- Long-term upward trend.
- Seasonal changes in passenger numbers.
- Non-stationary behaviour in the original data.
- Forecasted passenger values using different models.

The actual forecasting accuracy and model comparison should be evaluated using the generated results. 10. Conclusion

This project helped us understand the concepts and practical applications of Time Series Analysis using Python.

We analysed monthly airline passenger data from 1949 to 1960.

We studied trends, seasonality, stationarity, differencing, moving averages and time series decomposition.

We also implemented three forecasting techniques: ARIMA, Simple Exponential Smoothing and Holt-Winters.

The project helped us understand how historical data can be used to identify patterns and forecast future values.

Overall, this project improved our practical knowledge of Python libraries and time series forecasting techniques.

11. References

12. Python Documentation – https://docs.python.org/

13. Pandas Documentation – https://pandas.pydata.org/docs/

14. Matplotlib Documentation – https://matplotlib.org/

15. Statsmodels Documentation – https://www.statsmodels.org/

16. Scikit-learn Documentation – https://scikit-learn.org/

Project Title: Time Series Analysis Using Python

Dataset: Airline Passengers

Programming Language: Python